

3	(a)	(i)	8.0 cm		A1	[1]
		(ii)	$2\pi f = 220$ $f = 35$ (condone unit)		C1 A1	[2]
		(iii)	line drawn mid-way between AB and CD (allow ± 2 mm)		B1	[1]
		(iv)	$v = \omega a$ $= 220 \times 4.0$ $= 880 \text{ cm s}^{-1}$		C1 A1	[2]
	(b)	(i)	1. line drawn 3 cm above AB (allow ± 2 mm) 2. arrow pointing upwards		B1 B1	[1] [1]
		(ii)	1. line drawn 3 cm above AB (allow ± 2 mm) 2. arrow pointing downwards		B1 B1	[1] [1]
		(iii)	$v = \omega\sqrt{a^2 - x^2}$ $= 220 \times \sqrt{4.0^2 - 2.0^2}$ $= 760 \text{ cm s}^{-1}$ (incorrect value for x, 0/2 marks)		C1 A1	[2]